MIL-C-46697A (MU)
23 November 1966
SUPERSEDING
MIL-C-46697 (MU)
3 May 1963

MILITARY SPECIFICATIONS

CAP, SPIKE, WITH WASHER FOR CARTRIDGES
105MM, HEAT-T, M456A1 AND 90MM, HEAT-T, M431E1

1. SCOPE

1.1 This specification covers the spike cap and washer for use with cartridges, 105MM, HEAT-T, M456A1 and 90MM, HEAT-T, M431E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for.
MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Sampling by Attributes.

FSC 1315

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DRAWINGS

ARMY

8861463 - Cap, Spike.
8861464 - Washer, Cap.

PUBLICATIONS

ARMY

ET-8872553 - Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1. Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (dwg.) 8861463 and 8861464, as applicable, ar with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be of the proper finish and free of cracks, burrs; and chips, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.4 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such

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inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The

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first article sample shall consist of 25 complete sets of parts (i.e. 25 each of every component) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance of materials. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification,

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or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector.

4.3.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cap, spike (see dwg. 8861463)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 1.5 percent		
101.	Total length	Gage	01001
102.	Total inside depth	Gage	01002
103.	Depth to protrusion	Gage	01003
104.	Diameter of protrusion	Gage	01004
105.	Outside diameter	Gage	01005
106.	Concentricity of outside diameter with pitch diameter of thread, including perpendicularity with face of flange	Gage	01006
107.	Diameter of bore	Gage	01007
108.	Diameter of thread relief	Gage	01008
109.	Width of thread relief	Gage	01009

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110.	Depth from face of flange to bottom of thread relief	Gage	01010
111.	Excessive radius on outside corners	Gage	01011
112.	Radius at base of protrusion ..	Gage	01012
113.	Concentricity of flange diameter with pitch diameter of thread including perpendicularity with face of flange	Gage	01013
114.	Diameter of counterbore	Gage	01014
115.	Depth to bottom of counterbore.	Gage	01015
116.	Internal plating damaged	Visual	01016
117.	Pitch diameter of threads	Gage	01022
118.	Minor diameter of threads, maximum	Gage	01026
Minor:	AQL 2.5 percent		
201.	Diameter of flange	Gage	01017
202.	Thickness of flange	Gage	01018
203.	External plating damaged	Visual	01021
204.	Minor diameter of thread, minimum	Gage	01023
205.	Evidence of poor workmanship (see 3.3)	Visual	01024

4.3.2.2 Washer, cap (see dwg. 8861464)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Outside diameter	Gage	02001
102.	Thickness	Gage	02002
Minor:	AQL 1.0 percent		
201.	Inside diameter	Gage	02003
202.	Evidence of poor workmanship (see 3.3)	Visual	02004

4.3.3 Testing

4.3.3.1 Hardness.-Major defect - (Code No. 01025) - See dwg. 8861463 - One of the three following sampling plans shall be used depending on the type stock or the treatment process used.

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a) Furnace heat treatment. Hardness tests shall be performed on fifteen sample spike caps from each heat treat batch. If any sample fails to comply with the applicable requirements, the batch shall be rejected.

b) Pre-hardened stock. Hardness test samples shall be selected from each lot of spike caps using the sampling plan specified in MIL-STD-105 with an AQL of 1.0 percent to determine compliance with the applicable requirements.

c) Induction heat treatment. Hardness tests shall be performed on samples selected in accordance with MIL-STD-1235 using a CSP-1 plan having an AQL of 1.0 percent. Appropriate i and f values shall be specified by the procuring activity.

Testing shall be as specified in 4.4.1, using equipment in accordance with 4.3.4.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-8872553 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

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4.4.1 Hardness.-Hardness readings shall be taken at the periphery of the front face of the spike cap. When heat treating by induction, additional readings shall be taken on the side of the cap. Rockwell hardness tests shall be performed in accordance with Federal Test Method Standard No. 151, Method 243.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking

5.1.1 Level C.-Packing and marking shall be in accordance with Specification MIL-P-10025.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in

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the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item ofinspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other designs of inspection equipment shall be

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approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1315-A550

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